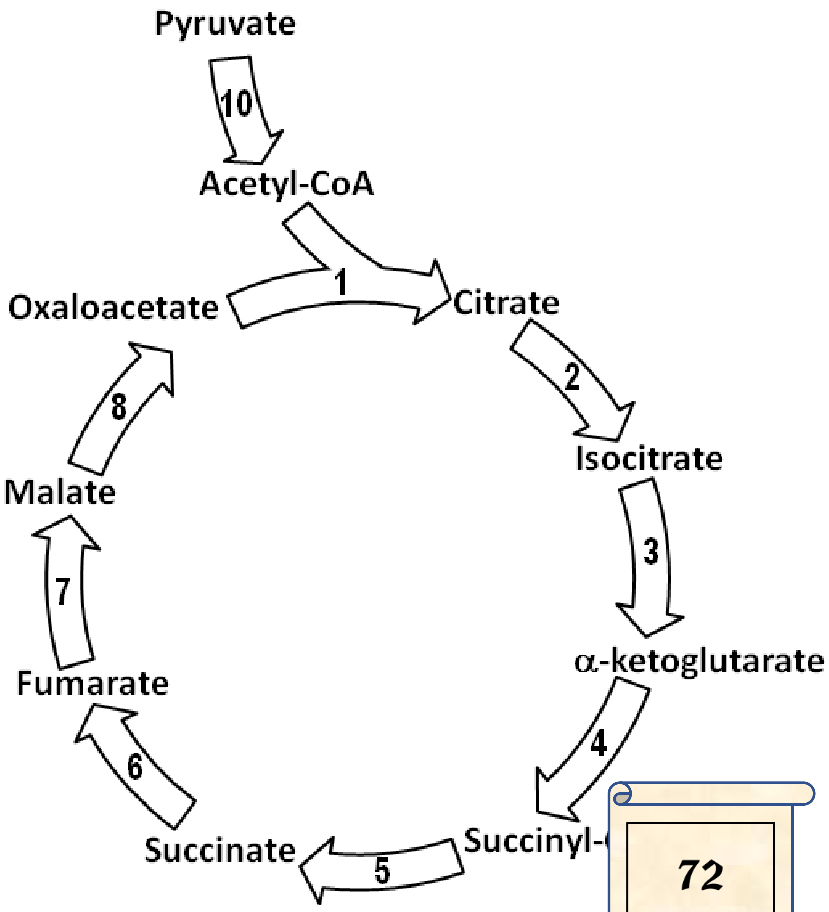
DM IMCQ#7 – KEY

F2020

Cumulative

1. A scientist studying metabolism of acetyl CoA, observes that it can enter a cyclic pathway and be converted to CO_2. Under controlled conditions, the intermediates of the pathway do not change with the introduction of the 2-carbon containing moiety of this molecule. Which of the following most likely must occur to permit metabolic flux through this cyclic pathway?

A. Enzyme 10 must be activated by Ca++	17%
B. Enzyme 4 must use biotin as coenzyme	17%
C. Enzyme 6 forms FADH2	17%
D. Enzyme 1 uses TPP and lipoic acid	17%
E. Enzyme 7 forms NADH	17%
F. CO2 is released in Step 8	17%



72

2. A 6-year-old girl is brought to the physician by his parents because of a 6-month history of poor feeding. Physical examination shows an irritable child with conjunctival pallor. The mother says that she likes to play toys painted with lead-containing paint. Laboratory tests show low hemoglobin levels. Which of the following enzymes is most likely inhibited in this child?

A. Acetyl CoA carboxylase	20%
B. MethylmalonylCoA mutase	20%
C. Bilirubin UDP-glucuronyl transferase	20%
D. Ferrochelatase	20%
E. Hydroxymethylbilane synthase	20%

3. A scientist is studying nitrogen metabolism and identifies an enzyme which catalyzes the exchange of the nitrogen from glutamate to oxaloacetate. Which of the following proteins is most likely responsible for this catalytic activity?

A. Glutaminase	17%
B. Glutamine synthetase	17%
C. Asparaginase	17%
D. Glutamate dehydrogenase	17%
E. Aspartate aminotransferase	17%
F. Alanine transaminase	17%

72

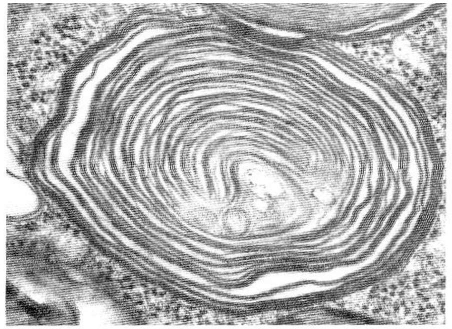
4. A 58-year-old man comes to the physician because of a 3-week history of yellow skin and sclera. He has a 10-year history of alcohol abuse. Physical examination shows scleral icterus. Liver biopsy shows tubulin damage. Which of the following molecules is most likely responsible for the biopsy findings?

A. Acetate	0%
B. Ketone bodies	0%
C. Ethanol in mitochondria	0%
D. Acetaldehyde	0%
E. VLDL	0%

72

5. A 2-year-old boy is brought to the physician because of a 3-month history of tripping over objects and developmental delay. Physical examination shows milestone developmental delay. Skin punch biopsy with abnormal lysosomes is shown. Which of the following compounds is most likely accumulating in his lysosomes?

A. Sphingomyelin	14%
B. Globoside	14%
C. Ganglioside	14%
D. Glycosaminoglycan	14%
E. Glycogen	14%
F. N-linked glycoprotein	14%
G. O-linked glycoprotein	14%



6. A 2-week-old female infant is brought to the physician by his mother because of a 4-hour history of alternating seizures and hypotonia. A provisional diagnosis of an inborn error of amino acid metabolism. Oral thiamine supplementation is prescribed which results in alleviation of the manifestations. Which of the following inherited enzyme deficiency disorders is most likely?

A. Phenylalanine hydroxylase
B. Homocysteine methyl transferase
C. Branched chain alpha ketoacid dehydrogenase
D. Homogentisate oxidase
E. Methylmalonyl CoA mutase
F. Medium chain acyl CoA dehydrogenase

0%
0%
0%
0%
0%
0%
0%

72

7. A 57-year-old man comes to the physician because of a 2-month history of productive cough and fever. Sputum culture is positive for tuberculosis. He is prescribed isoniazid. A vitamin is also prescribed to prevent the development of microcytic anemia. Which of the following enzymes most likely requires the coenzyme of this vitamin?

A. ALA synthase	20%
B. ALA dehydratase	20%
C. Porphobilinogen deaminase	20%
D. Uroporphyrinogen synthase	20%
E. Ferrochelatase	20%

72

8. A 30-year-old woman comes to the physician because of a 4-month history of fatigue. Physical examination shows pallor. Laboratory studies show megaloblastic anemia and homocysteinemia. Additional tests show excessive excretion of formimino-glutamate (FIGLU) following histidine ingestion. Which of the following vitamins is most likely deficient in this patient?

A. B6	0%
B. B12	0%
C. Biotin	0%
D. Folate	0%
E. Thiamine	0%

9. Twenty men aged 30- 35 years old are recruited for a study of the gastric phase of digestion. They are all given a standard meal and the changes in the stomach are observed when food enters the stomach. Which of the following events is most likely observed?

A. Chief cells release hydrochloric acid	20%
B. Activation of salivary amylase	20%
C. Activation of pepsinogen to pepsin	20%
D. Inhibition of lingual lipase	20%
E. Inhibition of gastric lipase	20%

10. A 20-year-old woman comes to the physician because she is planning to reduce her weight. Her BMI is 30kg/m2. Laboratory studies show elevated levels of a hormone that is produced by adipose tissue and controls appetite. Which of the following hormones is most likely measured in this patient?

A. Cholecystokinin	17%
B. Glucagon	17%
C. Leptin	17%
D. Ghrelin	17%
E. Insulin	17%
F. Adiponectin	17%

11. A 3-year-old boy is brought to the physician for a follow up examination because of a 4-month history of history of learning disability. Physical examination shows bilateral lens dislocation. Laboratory studies show elevated levels of a sulfur containing compound in his serum. Which of the following is most likely an additional complication of this disorder?

A.	Severe spinal arthritis	20%
B.	Premature vascular disease	20%
C.	Episodes of hyperammonemia	20%
D.	Episodes of hypoglycemia	20%
E.	Renal calcifications	20%

12. A 54-year-old man is brought to the emergency department because of sudden onset of severe pain in the abdomen. Laboratory studies show highly elevated serum lipase. Which of the following best describes the normal function of this enzyme?

A. Needs bile salts for its activity 20%
B. Degrades triacylglycerol to free glycerol 20%
C. Cleaves after lysine or arginine residues 20%
D. Is proteolytically activated 20%
E. Is secreted by the bile duct 20%

72

13. A research group is investigating the absorption of nutrients from the diet. In their experiments, the sodium-potassium ATPase in the mucosal cells is inhibited and nutrient absorption is compared to the uninhibited condition. Absorption of which of the following nutrients is most likely unaffected by inhibition of this transporter?

A. Hydrophobic amino acids	0%
B. Charged amino acids	0%
C. Glucose	0%
D. Fructose	0%
E. Galactose	0%

72

14. A 2-year-old girl is brought to physician by her parents because of a 1-month history of difficulty walking and progressive weakness in her legs. Laboratory studies show lysosomal accumulation of sulfatides. Which of the following is the most likely diagnosis?

A. Pompe disease	0%
B. Metachromatic leukodystrophy	0%
C. Fabry disease	0%
D. Gaucher disease	0%
E. I-cell disease	0%

15. A 58-year-old man is brought to the emergency department by his friend in an unconscious state. He has a 15-year history of chronic alcohol abuse. His friend says that he had large amounts of alcohol the previous night and did not eat any meals. Laboratory studies show hypoglycemia. High levels of which of the following is most likely found in the blood?

A. Lactate	20%
B. Glutamine	20%
C. Alanine	20%
D. Glycerol	20%
E. Glutamate	20%

16. Thirty 25-year-old men participate in a study of metabolic homeostasis during fasting. One group is fasting overnight, and the second group fasts for three days. Which of the following changes in blood levels are most likely found in individuals after 3 days of fasting when compared to individuals on an overnight fast?

A. The blood glucose level drops to about 2 mM	20%
B. Blood glucose is formed by glycogenolysis	20%
C. Blood levels of free fatty acids are lower	20%
D. Blood levels of ketone bodies are higher	20%
E. Insulin is absent in the blood	20%

72

17. A 2-year-old girl is brought to the emergency department because of a 6-hour history of lethargy. She has a 3-day history of fever and poor feeding. She had been diagnosed with MCAD deficiency by newborn screening. Which of the following is most likely observed during the metabolic crisis?

A.	Presence of ketone bodies in urine	0%
B.	Increased levels of very long chain fatty acids	0%
C.	Deficiency of peroxisomal fatty acid oxidation	0%
D.	Increased rates of fatty acid oxidation in liver	0%
E.	Impaired gluconeogenesis during fasting	0%

18. A 5-year-old boy is brought to the physician because of a 1-day history of pain in the right flank. Urinalysis shows crystals and RBCs. A provisional diagnosis of renal colic due to the passage of stones in his urine is made. Which of the following is most likely in this patient?

A. Defect in tryptophan transporter 0%
B. Defect in cystine reabsorption 0%
C. Defect in hepatic UDP-glucuronyl transferase 0%
D. Defect in homogentisate oxidase 0%
E. Deficiency of carbamyl phosphate synthetase-I 0%
F. Deficiency of ornithine transcarbamoylase 0%

19. A scientist is studying amino acid metabolism in experimental animals. She finds that a specific amino acid after deamination, produces both glucogenic and ketogenic substrates upon degradation. Metabolism of which of the following amino acids is most likely being studied?

A.	Aspartate	17%
B.	Leucine	17%
C.	Glutamate	17%
D.	Phenylalanine	17%
E.	Glycine	17%
F.	Glutamine	17%

72

20. A scientist is studying the incidence and distribution of inborn errors of metabolism. She finds that a specific disorder is more frequently observed in males. Which of the following enzyme deficiency disorders is most likely being studied by the scientist?

A.	Phenylalanine hydroxylase	13%
B.	Branched chain alpha ketoacid dehydrogenase	13%
C.	Ornithine transcarbamoylase	13%
D.	Methylmalonyl CoA mutase	13%
E.	Pyruvate kinase	13%
F.	Porphobilinogen deaminase	13%
G.	Carbamoyl phosphate synthase-1 (CPS1)	13%
H.	Carbamoyl phosphate synthase-2 (CPS2)	13%

21. A 60-year-old man comes to the physician because of a 2-month history of difficulty in walking and climbing stairs. Laboratory studies confirm adult onset Pompe disease. He is prescribed enzyme replacement therapy. Which of the following enzymes is most likely present in this infusion?

A.	$\alpha(1\rightarrow4)$ Glucosidase	20%
B.	Arylsulfatase A	20%
C.	β-Glucosidase	20%
D.	α-Galactosidase	20%
E.	Sphingomyelinase	20%

22. A researcher investigates hepatic ethanol metabolism with specific focus on MEOS and compares it to alcohol dehydrogenase (ADH). Which of the following best describes this system?

A. Uses NADH as coenzyme	20%
B. Has a higher affinity for ethanol than ADH	20%
C. Contains mainly CYP2E	20%
D. Forms acetate	20%
E. Found in mitochondria	20%

72

23. A 3-year-old girl is brought to the physician for a follow up examination because of repeated respiratory infections and steatorrhea. Her mother says that her sweat is very salty. A specific diet enriched with nutrients to improve her digestion is prescribed. Which of the following nutrients is most likely added to her diet?

A. TAGs with eicosapentaenoic acid 0%
B. Proteins found in whey 0%
C. TAGs with medium-chain fatty acids 0%
D. Proteins found in casein 0%
E. TAGs with dietary essential fatty acids 0%

24. A 2-year-old boy is brought to the physician by his parents because of a 2-week history of poor feeding. Physical examination shows an irritable child with conjunctival pallor. The mother says that he likes to play toys painted with lead-containing paint. Laboratory tests show low hemoglobin levels. Which of the following enzymes of heme synthesis is most likely inhibited in this child?

A. Uroporphyrinogen decarboxylase 0%
B. Uroporphyrinogen synthase 0%
C. ALA synthase 0%
D. Porphobilinogen synthase 0%
E. Hydroxymethylbilane synthase 0%

25. A research group is analyzing the conformational change that occurs when trypsinogen is activated to trypsin. They observe that a long narrow active site pocket is formed. Which of the following amino acids most likely binds to this pocket when a protein is cleaved by trypsin?

A. Methionine	0%
B. Asparagine	0%
C. Arginine	0%
D. Tyrosine	0%
E. Glutamine	0%

72

26. An 18-year-old man comes to the physician because of a 2-week history of skin lesions. He has been on a special diet containing raw egg white for the past 3 months. Avidin present in raw eggs inhibits absorption of biotin. Which of the following reactions is most likely impaired in this patient?

A.	Conversion of homocysteine to cystathionine	0%
B.	Conversion of methylmalonyl CoA to succinyl CoA	0%
C.	Synthesis of thymidine monophosphate	0%
D.	Conversion of phenylalanine to tyrosine	0%
E.	Synthesis of serotonin	0%
F.	Conversion of propionyl CoA to methylmalonyl CoA	0%

27. A 39-year-old woman comes to the physician because of a 3-week history of easy bruising. She has a 2-year history of steatorrhea. A provisional diagnosis of dietary fat-soluble vitamin deficiency is made. Which of the following is most likely required for efficient absorption of these nutrients?

A.	Glycochenodeoxycholic acid	20%
B.	Deoxycholic acid	20%
C.	Lithocholic acid	20%
D.	Urobilinogen	20%
E.	Stercobilinogen	20%

72

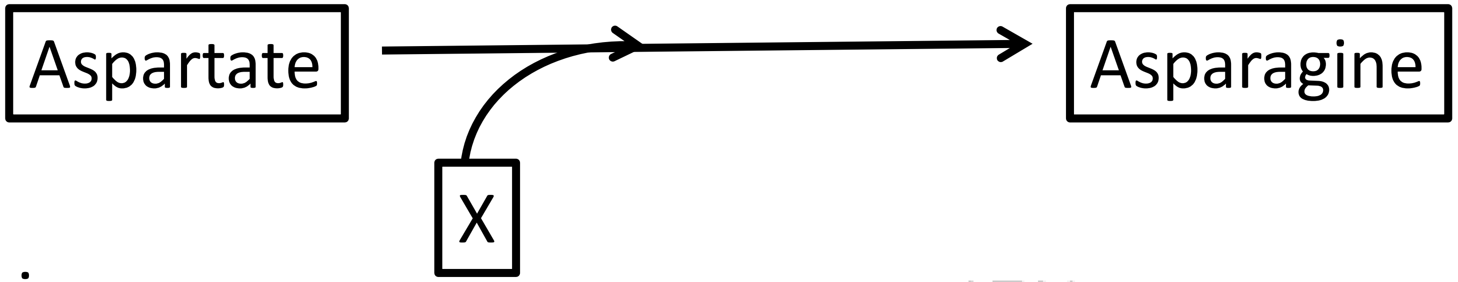
28. A scientist is studying amino acid metabolism in experimental animals. She finds that a specific amino acid after deamination, produces both glucogenic and ketogenic substrates upon degradation. Metabolism of which of the following amino acids is most likely being studied?

A.	Alanine	17%
B.	Asparagine	17%
C.	Lysine	17%
D.	Glutamine	17%
E.	Tyrosine	17%
F.	Glycine	17%

29. A scientist is studying aromatic amino acid metabolism. He observes that the first step in the degradation of phenylalanine is the formation of tyrosine, a different common amino acid. This reaction is catalyzed by phenylalanine hydroxylase and is dependent on the cofactor BH4. During the reaction, BH2 is formed. Which of the following molecules is most likely needed for the conversion of BH2 back to the active BH4 form?

A. Biotin	14%
B. ATP	14%
C. FADH2	14%
D. NADPH	14%
E. Phosphatidyl inositol	14%
F. cAMP	14%
G. Ca^{++}	14%

30. A scientist is studying nitrogen metabolism and observes that asparagine may be formed in a single step from aspartate. A figure of the reaction is shown. Which of the following molecules most likely represents compound X?



A. Glutamine	17%
B. NH3	17%
C. NADH	17%
D. NADPH	17%
E. Glutamate	17%
F. Alanine	17%

31. A 66-year-old man comes to the physician because of a 6-month-history of tingling of his extremities. Laboratory studies confirm vitamin B6 deficiency. Synthesis of which of the following molecules is most likely impaired as a result of this deficiency?

A. Methionine	0%
B. Homocysteine	0%
C. Cysteine	0%
D. Serine	0%
E. S-adenosyl methionine	0%

32. A 15-hour-old newborn boy is admitted to the NICU (neonatal intensive care unit) because of lethargy and seizures. Laboratory studies show hyperammonemia. Serum organic acid profile shows undetectable citrulline, and elevated orotic acid. Which of the following enzymes is most likely deficient in this patient?

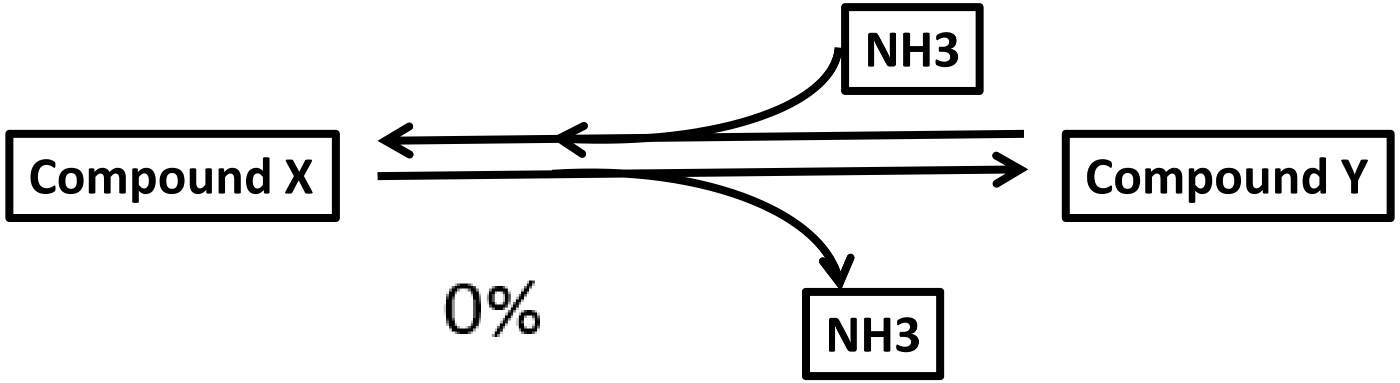
A. Argininosuccinate lyase	17%
B. Ornithine transcarbamoylase	17%
C. Carbamoyl phosphate synthetase-1	17%
D. Arginase	17%
E. Glutamate dehydrogenase	17%
F. Aspartate transaminase	17%

33. A 46-year-old woman comes to the physician because of a 3-month history of severe indigestion and diarrhea. Stool analysis shows unformed feces. Laboratory studies show decreased cholecystokinin release after food intake. Which of the following actions is most likely mediated by this hormone?

A. Release of trypsin from the pancreas	20%
B. Release of bile from the gallbladder	20%
C. Activation of prophospholipase A2	20%
D. Release of chyme into the duodenum	20%
E. Activation of MAG uptake into intestinal cells	20%

72

34. A scientist is studying nitrogen metabolism and identifies an enzyme that is, in part, regulated by the relative abundance of available substrates. A diagram of the reaction is shown. Which of the following enzymes most likely catalyzes this reaction?



A. Argininosuccinate lyase 0%
B. Ornithine transcarbamoylase 0%
C. Carbamoyl phosphate synthetase 1 0%
D. Arginase 0%
E. Glutamate dehydrogenase 0%
F. Aspartate transaminase 0%

35. A 6-year-old boy is brought to the physician by his parents for a follow up examination after being diagnosed with acute lymphoblastic leukemia (ALL) 6 months ago. He is prescribed an enzyme transfusion as part of his treatment regimen. Which of the following enzymes is most likely given to this patient?

A. Glutaminase	0%
B. Glutamine synthetase	0%
C. Asparaginase	0%
D. Glutamate dehydrogenase	0%
E. Aspartate transaminase	0%
F. Alanine transaminase	0%

36. A scientist is studying nitrogen metabolism in dogs in the fasting state. He withholds food from dogs for 24 hours. Following the fast, the dogs are fed a meal that contains solely 19 of the 20 standard amino acids. All of the animals develop a transient hyperammonemia spike. Which of the following amino acids is most likely missing from this diet?

A. Alanine	0%
B. Glutamine	0%
C. Glutamate	0%
D. Aspartate	0%
E. Arginine	0%
F. Cysteine	0%
G. Glycine	0%

Note that in this theoretical experiment, no animals would be harmed (except that they would be hungry for a day)

72

37. A 28-year-old man is brought to the emergency department by his friend in an unconscious state. He has a 5-year history of alcohol abuse. His friend says that he had large amounts of alcohol the previous night and did not eat any meals. Which of the following additional laboratory finding is most likely?

A.	Decreased plasma glucose levels	20%
B.	Decreased blood lactate levels	20%
C.	Increased NAPQI formation	20%
D.	Low hepatic NADH/NAD+ ratio	20%
E.	Increased oxalate levels	20%

72

38. A 3-year-old boy is brought to the physician for a follow up examination because of a 4-month history of history of learning disability. Physical examination shows bilateral lens dislocation. Laboratory studies show elevated levels of a sulfur containing compound in his serum. Which of the following enzyme deficiency disorders is most likely?

A. Homogentisate oxidase 20%
B. Cystathionine synthase 20%
C. Carbamoyl phosphate synthetase-1 (CPS-1) 20%
D. Phenylalanine hydroxylase 20%
E. Porphobilinogen deaminase 20%

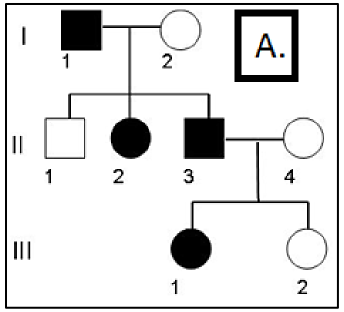
39. Thirty 25-year-old men participate in a study of metabolic homeostasis during fasting. Liver enzyme activities are measured on day 3 of fasting. Which of the following enzyme activities is most likely increased in the liver in these patients?

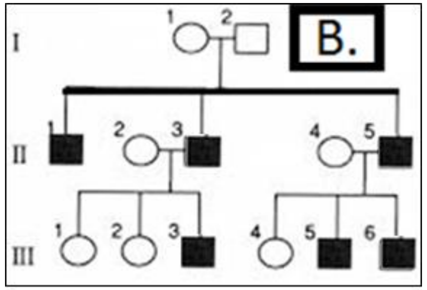
A. Phosphofructokinase-1 20%
B. HMG CoA reductase 20%
C. Acetyl CoA carboxylase 20%
D. Mitochondrial HMG CoA synthase 20%
E. Glycogen synthase 20%

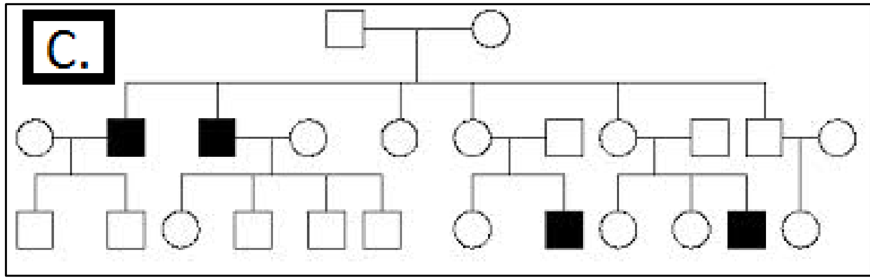
72

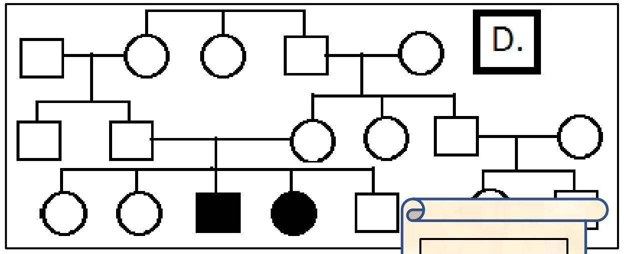
40. A 2-day-old newborn boy is brought to the emergency department because of a 6-hour history of alternating severe hypotonia and seizure. Laboratory studies show elevated blood ammonia levels. Urinalysis shows presence of orotic acid. Family history shows that other members of the family had previously been affected with the same disorder. Which of the pedigrees would most likely depict this family?

A. Pedigree A	17%
B. Pedigree B	17%
C. Pedigree C	17%
D. Pedigree D	17%
E. None of these pedigrees can describe this family	17%
F. Any of these pedigrees can describe this family	17%









72

Key

1. C
2. D
3. E
4. D
5. C
6. C
7. A
8. D
9. C
10. C
11. B
12. A
13. D
14. B
15. A
16. D
17. E
18. B
19. D
20. C
21. A
22. C
23. C
24. D
25. C
26. F
27. A
28. E
29. D
30. A
31. C
32. B
33. B
34. E
35. C
36. E
37. A
38. B
39. D
40. C

Thank you and all the best for your exams